



Fed-Ex

Project Description

Business Context FedEx Logistics manages a complex global supply chain, dealing with shipments across various regions, countries, and industries. This dataset provides an in-depth look at their logistics processes, capturing important information on purchase orders (POs), shipment methods, vendor agreements (INCO terms), delivery schedules, and product-specific details such as item descriptions and dosage forms. Effective management of these processes ensures timely delivery, minimizes freight costs, and improves customer satisfaction.

Given the rise of eCommerce and global distribution, companies like FedEx must continuously optimize their logistics operations to maintain a competitive edge. The dataset is designed to provide insights into how shipments are managed, identify bottlenecks or delays, and ensure cost-effectiveness. By analyzing the data, FedEx Logistics aims to streamline supply chain operations, improving delivery timelines and reducing costs for both the company and its customers.

Dataset Description

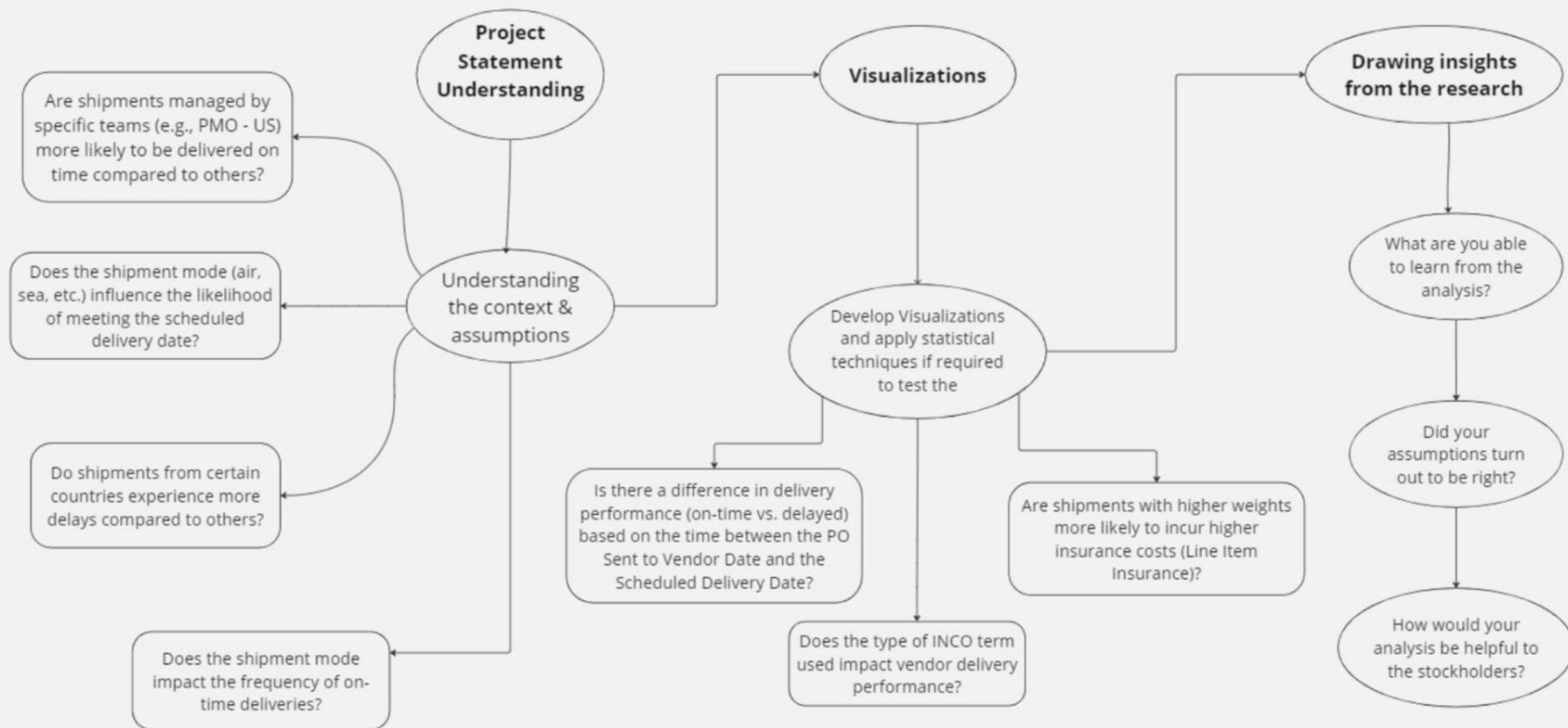
ID	Unique identifier for each logistics record.
Project Code	Code representing the project or initiative tied to the shipment.
PQ #	Price Quotation reference number.
PO / SO #	Purchase Order or Sales Order associated with the shipment.
ASN/DN #	Advanced Shipment Notice or Delivery Note reference.
Country	Destination country for the shipment (e.g., Côte d'Ivoire, Vietnam).
Managed By	Specifies which team manages the project (e.g., PMO - US).
<u>Fulfill Via</u>	Method of <u>fulfillment</u> , such as direct drop shipment.
Vendor <u>INCO Term</u>	Incoterms defining the shipment agreement (e.g., EXW, FCA).
Shipment Mode	The mode of transportation used (e.g., air).
PQ First Sent to Client Date	The date the Price Quotation was first sent to the client (some records indicate "Pre-PQ Process").
PO Sent to Vendor Date	<u>Date the</u> purchase order was sent to the vendor.
Scheduled Delivery Date	The planned date for delivery.

Brand	The brand name of the product (e.g., Reveal, Generic).
Dosage	The dosage strength of the pharmaceutical product (if applicable).
Dosage Form	The form in which the product is delivered (e.g., test kit, oral suspension).
Unit of Measure (Per Pack)	The number of units per pack.
Line Item Quantity	The total number of units shipped.
Line Item Value	The total value of the line item in the shipment.
Pack Price	Price per pack of the item.
Unit Price	Price per individual unit of the item.
Manufacturing Site	The location where the item was manufactured (e.g., Ranbaxy Fine Chemicals LTD, Aurobindo Unit III, India).
First Line Designation	Indicates whether the product is designated for first-line use.
Weight (Kilograms)	The weight of the shipment.
Freight Cost (USD)	The cost of shipping the item in U.S. dollars.
Line Item Insurance (USD)	Insurance cost for the line item, if applicable.

Main Libraries to be used:

- **Pandas:** For data manipulation, cleaning, and analysis.
- **NumPy:** For efficient numerical computations and array handling.
- **Matplotlib:** For creating basic static visualizations like line and bar charts.
- **Seaborn:** For advanced statistical visualizations and better aesthetics.
- **Plotly:** For interactive and dynamic data visualizations.

Project Architecture:



Project Evaluation Criteria

Rubrics	Weightage
Summary and Technical Documentation in Collab Notebook	10
Exploration, Namely: Head, Tail, Summary, Data Dictionary	5
Looking for and Handling NaN/ Null/ Missing Values	5
Trying to Get Some Conclusion From Data, Correlation, Trends	10
Accomplish Various Milestones Given in the Problem Statement	10
Using Visualization (At least Five Different Types) for Presenting the EDA	15
Final Summary of Conclusion	5
Commented Code	5
Proper Output Formatting	5
Modularity of Code	5
Video Presentation	20
Fluency and Grammatical Accuracy in Video	5